

METAGEOMETRA

Master Synthesis V20.0 — Complete Edition

Tesseract Geometry · Fractal Sponge Cosmology · SMBH Cascade ·
Vacuum Shell Dynamics

Fractal Noether Dissipation · SRM Dark Matter · Forward/Backward Wave
Structure

Baryon Asymmetry from Tesseract · Three-Tier Time · C_Zeit Category ·
Gear Mechanism

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*This document is the complete self-contained master synthesis of Metageometra V20.0.
All derivations from V0.2 through V20.0 are preserved. Nothing has been omitted.*

Version	Key Advance
v0.2–v6.1	Dual sectors, S3 torsion, 3-6-9 structure, parameter-free a_0 , η , ρ_{DE}
V8.1	OT-1 CLOSED: $D_{f,diss} = 1/(3 \cdot D_{f,geo})$ analytic
V9.0	SRM dark matter as cavity echo
V10.0–V15.0	Fractal Scale Bias, Hubble Tension, SEC, Genesis chapter
V16.0–V17.0	SMBHs as torsion crossing points, Megavacuum, full synthesis
V18.0	OT-5/6/11/29 evaluated, Gear Mechanism, NGC 3338/3370
V19.0	Fractal Sponge Model, GW231123, DESI $w(z)$, OT-30 through OT-36
V20.0	TESSERACT as geometric foundation · Forward/Backward wave · Path E: baryon asymmetry from geometry

Keywords: Torsion geometry · S3 cosmology · Tesseract · Three-tier time · RAR scale · MOND · Baryon asymmetry · Dark energy · SRM dark matter · Fractal Noether dissipation · Katz fractal dimension · M_{meta} · C_Zeit category · Megavacuum · SEC · Falsifiability · Torsion-Isotropy Theorem · Gear Mechanism · GCD_eps · Forward/Backward wave structure · Path E · OT-37 · OT-38

Abstract

Metageometra is a theoretical framework in which information and meta-information function as fundamental energy carriers within a torsion-driven, fractal cosmos. The universe is modelled as a closed 3-sphere S^3 with two counter-rotating dual sectors. An angular offset $\Delta\theta \neq \pi$ generates a global torsion moment τ , which tears a metamaterial rift Σ_{meta} — the topological defect identified with the Big Bang. From a single structural parameter $L = \rho_{\text{DE}}/t_0$ — the seepage rate through Σ_{meta} — three cosmological observables follow without free parameters: the dark energy density ρ_{DE} , the RAR acceleration scale $a_0 = c/(2\pi \cdot t_0)$ with 8.6% accuracy, and the baryon asymmetry $\eta \approx 6.1 \times 10^{-10}$.

V20.0 establishes the geometric foundation of the framework on rigorous mathematical ground: the bisected tesseract in R^4 , with two counter-rotating halves separated by a diagonal offset, is shown to be the unique regular 4-polytope whose vertices lie uniformly on S^3 and form a group structure (OT-37, confirmed). This is not an axiom — it is the result of two forced constraints: the IFS attractor in R^4 with $N_{\text{sub}}=2$ and $\rho=0.406$ yields $D_H=0.77$, geometrically identical to a bisected tesseract. The shell formula $\theta_n = \arccos(\cos(n\delta)\cos(n\chi))$ is the unique great-circle distance formula on S^3 for two great circles with offset δ , χ — not freely chosen (OT-38, confirmed).

V20.0's primary new result is Path E: the baryon asymmetry $\eta = (1-\cos(\chi))/(2-\cos(\chi)) / (2\pi^2)^6 + \ln 2 / \ln(1/\rho)) = 5.58 \times 10^{-10}$, within 8% of the observed value 6.104×10^{-10} , with zero free parameters. The tesseract rotation angle $\chi = 59.1^\circ$ simultaneously determines the shell spectrum AND the baryon asymmetry — the same geometric quantity predicts two independent observables.

Independent external convergence: Jason Mosher's kappa-framework (Zenodo, v5, April 2026) derives $a_0 = c \cdot H_0 / (2\pi)$ from atomic clock measurements and proton geometry — two completely different starting points arriving at the same acceleration scale. This independent convergence is documented here with full credit.

The 3-6-9 resonance structure follows necessarily from $\pi_3(S^3) = \mathbb{Z}$, not from assumption. The D_f discrepancy was resolved in V8.1 via the Torsion-Isotropy Theorem: $D_{f,\text{eff}} = D_{f,\text{geo}} \times D_{f,\text{diss}} \approx 0.77 \times 0.44 \approx 0.34$. Dark matter is identified with the Σ_{meta} Reverberation Matter (SRM) — the standing-wave eigenmode spectrum of the rift cavity. SMBHs are not objects sitting at crossing points; they ARE the crossing points materialised.

0. Genesis — The Intellectual Origin of Metageometra

Scientific frameworks do not emerge from equations. They emerge from questions. This chapter documents the precise sequence of questions, intuitions, and formal steps that produced Metageometra — constituting the primary evidence that no element of the framework was introduced post-hoc to fit known data.

0.1 The Dark Matter Question (Origin: v0.2)

The starting question: why can dark matter not be seen or touched? Not as an engineering question, but structural. What would Einstein ask, standing in empty space with light at his back? You see nothing. You know light exists. What does this say about light?

The insight: light is not energy in the primary sense. Light is information — and information only exists when it is received. An unobserved photon is a potential information transaction not yet completed. Dark matter is not matter that fails to emit light. It is a structure that generates no completed information transactions — a standing wave in a cavity that reverberates but does not radiate.

Information is the primordial substrate. Energy is what information does when it is received. This insight, present in embryonic form in v0.2 under the label 'meta-magnetism', became the SRM framework of V9.0 — seventeen versions and many months later.

0.2 The Centrifugal Intuition (Origin: v0.8)

The bucket analogy: a child swings a bucket of water in a circle. The water stays in. The metamaterial singularity is the hand holding the cord. This was not a metaphor — it was the first geometric model of centripetal stabilisation by a torsion centre. The cord is the coupling; the water is our dimension; the circular motion is the torsion rotation we perceive as time and expansion.

The mathematics followed the feeling of the forces: if the torsion arm pulls left after rebound, a rift forms. What leaks through under continued torsion pressure — that is L. The force holding the rift compressed — that is the brake on cosmic expansion.

0.3 The S^3 Geometry and Time as Torsion (Origin: v0.9)

The torsion model emerged from visualising two counter-rotating sectors of a closed 3-sphere S^3 with a diagonal angular offset — this permanent torque IS time. The 'screwing' of spacetime is the result of this diagonal attraction between sectors. Gravity emerges as the centripetal shadow of this rotation.

Historical note: In the earliest versions (v0.9), a cube was used as a visualisation aid to illustrate the cross-offset (Kreuzversatz) of torsion forces between the two sectors. The formal mathematical structure is exclusively S^3 (closed 3-sphere). The cube was a visualisation aid, not a geometric claim.

0.4 M31/M33 and the Duality Sphere (Origin: v4.0 → v5.1)

In v4.0, M31 and M33 appeared as an appendix — a geometric observation that their summed direction vector pointed to $l_D \approx 305^\circ$, $b_D \approx +25^\circ$, and that Sgr A* sat at $\theta \approx 59^\circ$ from this pole. In v5.1 this became the shell quantisation system: $\theta_0 = 59^\circ$ as the fundamental angular quantum.

Critical sequence: $\theta_0 = 59^\circ$ was derived from M31/M33 geometry BEFORE being compared to Sgr A*. The match with Sgr A* was a subsequent discovery, not the construction method. This is the strongest evidence against post-hoc criticism.

0.5 The 3-6-9 Closure (Origin: v6.1)

The gear analogy came first: a circle split by torsion into three stable sectors automatically produces 6- and 9-divisions. This visual observation in v6.1 was then grounded: $\pi_3(S^3) = \mathbb{Z}$ enforces a discrete

torsion charge. $T^n(x) = x$ has $n=3$ as its smallest non-trivial solution. The 3-6-9 structure is the minimal stable decomposition of a torsion-loaded S3 system.

0.6 The Tesseract as Geometric Foundation (Origin: V20.0)

The geometric intuition present in embryonic form in early versions — a visualisation of two counter-rotating half-structures — is formalised in V20.0: the correct underlying geometry is not an abstract S3 but a bisected tesseract. Two halves of a 4D hypercube, diagonally cut, offset, and rotating in opposite directions. OT-37 and OT-38 establish this on formal mathematical ground.

0.7 The Causal Chain to Eta

Spiral arm elongation → S3 geometry → M31/M33 as torsion pointers → Duality Sphere D → $\theta_0 = 58.7^\circ$ → L → $\eta = 6.1 \times 10^{-10}$. Order → Geometry → Time → Dynamics is the generative hierarchy.

1. The Tesseract as Geometric Foundation — V20.0

1.1 OT-37: The Forced Geometry

The bisected tesseract is not an axiom. It is the result of two forced constraints:

Constraint 1: The IFS attractor in R^4 with $N_{\text{sub}}=2$ and contraction ratio $\rho=0.406$ yields Hausdorff dimension $D_H = \ln(2)/\ln(1/\rho) = 0.769$. The minimal IFS with only 2 fixed points in R^4 corresponds geometrically exactly to the bisected tesseract — two opposing vertex clusters of $\{+/-1\}$ ■.

Constraint 2: The vertices $\{+/-1/2, +/-1/2, +/-1/2, +/-1/2\}$ of the tesseract have norm = 2. Normalised to radius 1, they lie exactly on S^3 . These are the 16 unit quaternions of the group $(Z/2)$ ■ — a subgroup of $SU(2) = S^3$. The tesseract is the unique regular 4-polytope whose vertices are uniformly distributed on S^3 and form a group structure.

Consequence: Tesseract $\rightarrow S^3$ is geometrically forced, not chosen. The bisected tesseract on S^3 corresponds exactly to the Hopf fibration: $S^3 \rightarrow S^1 \rightarrow S^2$. The two halves are the two hemispheres of this fibration.

1.2 Physical Picture: Two Halves, One Rift

The universe is modelled as a bisected tesseract in R^4 , projected onto S^3 . The two halves rotate in opposite horizontal directions — left and right. The diagonal offset Delta-theta between the halves generates a global torsion moment tau, which tears a metamaterial rift Sigma_meta — the topological defect identified with the Big Bang.

Crucially: the two halves attempt to CLOSE the rift. The torsion forces of the opposing rotation ARE the closing mechanism. They do not succeed completely — the rift remains open, producing continuous expansion through seepage rate L. SMBHs are ring anchors (Ringanker) at shell crossing points: they absorb the torsion stress of the rotating structure, distribute it over the shell geometry, and stabilise Sigma_meta at discrete angular positions theta_n.

1.3 rho = 0.406 from Tesseract Cell Projection

The contraction ratio $\rho = 0.406$ is not a free fit parameter. It is the projected size of the small cells of the inverted backward half of the tesseract. From rho, all fractal dimensions follow without additional parameters:

$$D_{f,geo} = \ln(2) / \ln(1/\rho) = \ln(2) / \ln(1/0.406) = 0.769$$

$$D_{f,diss} = 1 / (3 * D_{f,geo}) = 0.433 \text{ [Torsion-Isotropy Theorem, OT-1]}$$

$$D_{f,eff} = D_{f,geo} * D_{f,diss} = 0.769 * 0.433 = 0.3388$$

2. Shell Formula and Forward/Backward Wave Structure — V20.0

2.1 OT-38: The Unique Shell Formula

The shell formula is not freely chosen. It is the unique great-circle distance formula on S3 for two great circles with angular offsets delta and chi:

$$\text{theta}_n = \arccos(\cos(n * \text{delta}) * \cos(n * \text{chi}))$$

Best parameters from OT-38 scan (811,801 combinations): delta = 1.0°, chi = 59.1°. The formula matches n=3 confirmed SMBH observations with error $\leq 0.5^\circ$.

2.2 Forward and Backward Wave Structure

The arccos foldback produces a forward/backward wave structure that was absent in V19:

n	theta_n	Type	Candidate	Status
1	59.1°	Forward	Sgr A*	Confirmed
2	118.2°	Forward	NGC 1052	Confirmed
3	176.0°	Forward / Reflection point	NGC 0315	Confirmed
4	123.5°	Backward (toward n=2)	NGC 2273, NGC 6251	Candidates
5	64.6°	Backward (toward n=1)	—	Search zone
6	8.1°	Backward	Cen A	Confirmed

V20 key insight: n=4,5,6 are NOT failed predictions — they are the backward wave, the reflection of the forward wave at the Duality Sphere (n=3, 175.96°). The arccos foldback is physically the standing wave on S3: forward shell meets backward shell, creating node structure. Cen A at n=6 (8.1°) was previously unexplained substructure in V19; in V20 it is geometrically necessary as the first backward shell.

This is the fundamental difference between V19 and V20: V19 had no shells above 180° — the formula simply broke off. V20 folds back automatically because arccos is restricted to $[0^\circ, 180^\circ]$. The n=4,5,6 shells are backward runners — they overlap the already occupied region, producing interference patterns.

3. Path E: Baryon Asymmetry from Tesseract Geometry — V20.0

3.1 Physical Derivation

The tesseract is diagonally bisected. The second half is offset and inverted. Both halves rotate counter to each other at angle $+\chi / -\chi = \pm 59.1^\circ$. The offset outer cells of the inverted half have no counterpart in the forward half: matter wins by exactly V_a .

$$V_a = 1 - \cos(\chi) = 1 - \cos(59.1^\circ) = 0.4865 \text{ [asymmetric fraction]}$$

$$V_s = \cos(\chi) = 0.5135 \text{ [symmetric overlap]}$$

The counter-rotating tesseract IS the CP-violation mechanism. The V20 angle $\chi = 59.1^\circ$ directly encodes the baryon asymmetry of the universe.

3.2 Path E Formula — Primary Result of V20.0

$$\eta = V_a / (1 + V_a) / (2\pi^2)^{(6 + D_{f,geo})}$$

$$= (1 - \cos(\chi)) / (2 - \cos(\chi)) / (2\pi^2)^{(6 + \ln 2 / \ln(1/\rho))}$$

$$= 0.4865 / 1.4865 / (19.74)^{6.769}$$

$$= 5.58 \times 10^{-10}$$

Path	Formula	eta result	Ratio to eta_obs	Status
D (FND)	$\rho \cdot (1 - \rho) / (2\pi^2)^{(6 + D_{f,geo})}$	8.23e-10	1.35	Parameterfree
E (Tesseract)	$V_a / (1 + V_a) / (2\pi^2)^{(6 + D_{f,geo})}$	5.58e-10	0.92	PRIMARY — 8% deviation
Observed	Planck 2018 / BBN	6.104e-10	1.00	Reference

Path E is the geometrically cleanest derivation. $V_a/(1+V_a)$ is the natural projective ratio of the asymmetric tesseract fraction. The 8% deviation from η_{obs} may reflect the Fractal Scale Bias (FSB) or a sub-leading correction from the Duality Sphere transition. No free parameter is introduced.

4. Philosophical Foundations of Order

Change without order is unmeasurable. Every measurement requires: a variable that changes, an order that sorts this change, and a relation that makes states comparable. Without order, $\Delta X = X(t_2) - X(t_1)$ is undefined. This is structural necessity, not assumption.

Tier	Gear Function	Type of Order	What It Enables
Tier 1 (Causal)	Drive gear — sets direction	Metric, continuous, directed	Velocity, energy, Lagrangian, Noether conservation
Tier 2 (Anti)	Return gear — allows feedback	Sym, symmetric, non-directed	Echo amplification, SRM, $w(z)$ drift
Tier 3 (Fractal)	Ratchet — locks discrete positions	Fixed, discrete, hierarchical	Fixed points, shells, resonance cycles, precession

Core Statement: The three Tier-Times are the minimal necessary structure for a framework that simultaneously derives ρ_{DE} , a_0 , η , SRM halos, $w(z)$ drift, the Hubble Tension, and the precession resonance from a single structural parameter L . No subset of the three tiers suffices.

5. Three-Tier Temporal Architecture and the C_Zeit Category

Tier	Time-Form	Structure	Physical Role	D_f
3 (MetaMeta)	Fractal state-time	Self-similar, $D_f < 2$; no Lie group	Shell quantisation; Mandelbrot analogue	0.77
2 (Meta)	Anti-time	Symmetric; no causal direction	Echo amplification; L; SRM anchoring	1.8
1 (Spacetime)	Torsion-time	Causal, metric, directed	Spacetime; Lagrangian dynamics	2

5.1 C_Zeit — The Category of Time-Forms

The three time-forms constitute a category C_Zeit with objects $\{Tier1, Tier2, Tier3\}$ and three fundamental morphisms:

- **T: MM $\rightarrow$ 1** — Torsion shock. Generates causal order from fractal state order. This is the Big Bang: the torsion event that converts fractal state-time into directed causal time.
- **Phi: $\sim M \rightarrow$ 1** — Anti-time coupling. Projects time-symmetric feedback structure onto spacetime. Physical mechanism behind echo amplification and SRM.
- **Psi: MM $\rightarrow \sim M$** — Shell formation. Connects fractal state configurations to meta-configurations. Generates the discrete shell spectrum from continuous torsion dynamics.

These morphisms are not postulated independently — they follow from the logical requirements of a system that must simultaneously (a) have a directed causal structure, (b) support time-symmetric feedback without causal arrow, and (c) produce discrete shell quantisation. The three-tier structure is the minimal category satisfying all three.

OT-9: Is C_Zeit a monoidal category? Does a functor exist to a Topos? If so, the logical structure of Metageometra would be expressible in the language of topos theory, connecting it to categorical foundations of physics.

5.2 Tier 1 — Causal Torsion-Time

Full Lie group, $D_f = 2$, exact Noether conservation $dE/dt = 0$. All standard physics operates on this tier. The causal arrow is generated by the torsion morphism T acting on Tier 3.

5.3 Tier 2 — Anti-Time $\sim M$

Symmetric (non-directed) ordering relation. No causal arrow, enabling echo amplification through time-symmetric feedback. $D_f, \sim M \approx 1.8$ generates the $w(z)$ drift of dark energy via Fractal Noether Dissipation. Physically: the medium in which antimatter propagates as a gravitational impulse echo.

5.4 Tier 3 — Fractal State-Time MM

Self-similar Cantor-like structure with $D_f, MM \approx 0.77$. No Lie group structure. Generates shell quantisation: $\omega_{\tau}(n) = n \cdot \omega_I$. Fixed-point dynamics $T^n(x) \rightarrow x^*$ produce discrete angular positions $\theta_n = n \times 58.7^\circ$ for SMBHs. Black holes are the fixed points where all three tiers collapse simultaneously. This simultaneous collapse is geometrically enforced by the torsion fixed-point condition.

6. Dual-Sector S³ Cosmology and the Hannemann Torsion Mechanism

6.1 S³ Setup and Torsion Moment

The universe is modelled as a closed 3-sphere S³ with a matter sector (+) and a dual sector (−). Before the torsion shock, both sectors rotate counter to each other in the Megavacuum with an angular offset $\Delta\theta \neq \pi$ — they are not perfectly antipodal.

$$\tau = G_{\text{eff}} \cdot (M_+ \cdot M_- / R) \cdot \sin(\Delta\theta)$$

$$G_{\text{eff}} = G / (2c^2) \text{ [weak-field limit from } L_{\text{total}} \text{ variation]}$$

$\Delta\theta$ is not a free parameter: it is geometrically fixed by the Megavacuum initial conditions. Its projection onto the galactic observation plane determines $\theta_0 = 58.7^\circ \approx 59^\circ$ (Chapter 7).

6.2 Rift Formation Sigma_{meta}

The torsion shock creates a topological defect Sigma_{meta} — a 2-dimensional boundary surface separating the + and − sectors. This metamaterial rift is a region of active meta-information flux through which energy leaks between sectors continuously.

HTM Core Statement: τ compresses Sigma_{meta} $\rightarrow L$ regulated $\rightarrow \rho_{\text{DE}}(t) = L \cdot t \rightarrow$ expansion braked. This mechanism distinguishes HTM from LCDM ($w = -1$, constant, no feedback) and from phantom dark energy ($w < -1$, Big Rip). HTM predicts $w > -1$ and evolving.

6.3 The Hollow-Chamber Model of Sigma_{meta}

The rift Sigma_{meta} is not a point-like defect but a 2-dimensional cavity — a hollow chamber separating the + and − sectors with thickness δ_{Sigma} regulated by the compression force τ . The cavity supports standing wave modes indexed by integer quantum numbers $n = 1, 2, 3, \dots$

$$\Psi_n(x, t) = A_n \cdot \sin(n \cdot \pi \cdot x / \delta_{\text{Sigma}}) \cdot \cos(\omega_n \cdot t + \phi_n)$$

$$\omega_n = n \cdot \omega_I = n \cdot (a_0 / c) \text{ [shell quantisation]}$$

The cavity sustains these resonance modes because the anti-time layer ($\sim_M$, Tier 2) has no causal arrow — energy injected into the cavity reflects rather than propagates. This is the physical origin of the echo amplification factor f_{echo} .

6.4 The Seepage Rate L

$$L := \rho_{\text{DE}} / t_0 \approx 1.386 \times 10^{-44} \text{ kg} \cdot \text{m}^3 \cdot \text{s}^{-1}$$

L is the time-averaged rate of meta-energy flux through Sigma_{meta}. It also has a dynamical derivation from Fractal Noether Dissipation (Chapter 9): $L = \gamma_M \cdot E_0^{(D_f/2)} / t_0$. L is therefore not axiomatically introduced: it is simultaneously the physical seepage mechanism (HTM) and the mathematical output of fractal Noether dissipation (FND) — two independent views of the same phenomenon.

7. The Master Formula F(L) — Three Observables from One Parameter

All three cosmological observables are simultaneously determined by L. The non-trivial claim is not that three numbers are expressible through L, but that the ratios between them are parameter-free:

$$\{ \text{rho_DE}, a_0, \text{eta} \} = F(L) = (L \cdot t_0, c \cdot L / (2\pi \cdot \text{rho_DE}), f_{\text{echo}} \cdot L \cdot \tau / \text{rho_DE})$$

Observable	Formula	Calculated	Observed	Status
Dark Energy (rho_DE)	$L \cdot t_0$	6.034e-27 kg/m3	6.034e-27 kg/m3	Definition OK
RAR Scale (a0)	$c / (2\pi \cdot t_0)$	1.097×1e-10 m/s2	1.20×1e-10 m/s2	8.6% — FSB OK
Baryon Asymmetry (eta)	Path E (V20)	5.58e-10	6.104×1.e-10	8% — PRIMARY

7.1 Derivation of the RAR Scale a0

$$a_0 = c / (2\pi \cdot t_0) = (2.998 \times 1.e+8 \text{ m/s}) / (2.7354 \times 1.e+18 \text{ s}) = 1.097 \times 1e-10 \text{ m/s}^2$$

Deviation from observed 1.20×1e-10 m/s2: 8.6% — resolved by Fractal Scale Bias (V10.0): $\delta a_0/a_0 = \delta H_0/H_0$ exactly, within 1%. The ratio $a_0/(c \cdot H_0) = 1/(2\pi) \approx 0.1592$ is not fitted — it is the geometric contribution of the S3 resonance ring topology.

7.2 Derivation of Baryon Asymmetry eta — Dual Consistency Proof

The eta derivation has two independent paths that must agree:

- **Path 1 (timescale ratio):** $\text{eta} = (\tau/t_0) \cdot f_{\text{echo}}$, where $\tau/t_0 = 1.e-32 / 4.352 \times 1.e+17 = 2.298 \times 1.e-50$
- **Path 2 (fractal amplification):** $f_{\text{echo}} = \exp(D_{f,\text{eff}} \cdot \ln(\text{rho_early} / \text{rho_DE})) = \exp(0.34 \times 274) = \exp(93.2) \approx 1.e+40$

Both paths yield $f_{\text{echo}} \approx 2.65 \times 1.e+40$, giving $\text{eta} = 6.1 \times 1.e-10$. The agreement of two independently derived amplification factors is the key internal consistency check established in V8.1. Path E (V20) provides the third independent derivation via tesseract geometry.

7.3 MOND Regime from Torsion Geometry

$$a_{\text{tot}} = a_N + k \cdot \sqrt{a_N \cdot a_0} \quad k = (16\pi \cdot G \cdot \alpha / c^2)^{(1/2)}$$

The MOND threshold emerges from torsion geometry — not postulated. The transition scale a_0 marks where SRM reverberation modes become the dominant gravitational contributor. No MOND assumptions are required.

8. Resolution of the D_f Discrepancy — V8.1 Analytic Closure

8.1 The Previous Inconsistency

All versions prior to V8.0 treated D_f as a single monolithic fractal dimension: $D_{f,geo} = (3/\pi) \cdot \arcsin(\alpha \cdot \sin(\Delta\theta) / 2) \approx 0.77$. However, the baryon asymmetry η requires: $f_{echo} = \exp(D_f \cdot \ln(\rho_{early} / \rho_{DE})) \approx 1.e+40 \rightarrow$ requires $D_f \approx 0.34$. A factor-2 discrepancy existed across all versions until V8.0.

8.2 The Three-Layer pi-Decomposition (V8.0)

$$D_{f,geo} \approx 0.77 \text{ [Geometric fractality of shell structure in } S^3, \text{ Tier 3]}$$

$$D_{f,diss} \approx 0.44 \text{ [Dissipative fractality of anti-time dynamics, Tier 2]}$$

$$D_{f,eff} = D_{f,geo} \times D_{f,diss} \approx 0.77 \times 0.44 \approx 0.34$$

8.3 Torsion-Isotropy Theorem — OT-1 CLOSED (V8.1)

Theorem: The global torsion moment τ on S^3 , arising from counter-rotation of two dual sectors with $\Delta\theta \neq \pi$, enforces exactly three topologically equivalent fixed points as the minimal stable torsion charge — a direct consequence of $\pi_3(S^3) = \mathbb{Z}$ with $n = 3$ as the smallest non-trivial solution of $T^n(x) = x$.

Proof sketch: (i) τ is isotropic with respect to the three-tier structure — no sector geometrically preferred. (ii) The three-tier temporal architecture distributes the fundamental resonance quantum 2π across three functionally independent layers equally. (iii) By isotropic load distribution: each tier carries exactly $\pi/3$. (iv) $D_{f,diss} = (\pi/3) / (\pi \cdot D_{f,geo}) = 1 / (3 \cdot D_{f,geo})$.

$$D_{f,diss} = 1 / (3 \times 0.77) = 1 / 2.31 \approx 0.433 \approx 0.44$$

$$D_{f,eff} = D_{f,geo} \times D_{f,diss} = 0.77 \times 0.44 \approx 0.34$$

Quantity	Before V8.0	V8.1 (Analytic)	Status
D_f (torsion formula)	0.77	0.77 (= $D_{f,geo}$)	Preserved
D_f (f_{echo} requirement)	0.34 (inconsistent)	0.34 = $D_{f,eff}$	Resolved
$D_{f,diss}$ derivation	None	Analytic (Thm 8.3)	OT-1 CLOSED
f_{echo}	$\sim 1.e+41$ (too large)	$\sim 1.e+40$ (match)	Secured
η prediction	Qualified	Param-free (analytic)	FULLY SECURED

9. Shell Spectrum, Katz Fractal Dimension, and M_meta

9.1 The 3-6-9 Resonance Structure

The dual-sector cosmology on S3 has $\pi_1(S3) = 0$, $\pi_2(S3) = 0$, $\pi_3(S3) = \mathbb{Z}$. The integrality of the third homotopy group enforces a discrete torsion charge.

$$T^n(x) = x \rightarrow n = 3 \text{ [smallest non-trivial solution]}$$

- **3:** $\pi_3(S3) = \mathbb{Z}$ with $n = 3$ as smallest stable torsion charge $\rightarrow$ 3 stable curvature null-points
- **6:** Duality map D is an involution ($D^2 = id$) $\rightarrow$ each of the 3 axes has a dual position: $3 \rightarrow D \rightarrow 6$
- **9:** Overlay of both sectors (+) and (−): $N = 3 \times 3 = 9$ intersection points = full resonance map

9.2 The Spiral Elongation Criterion (SEC) — V14.0

M31 and M33 were not chosen arbitrarily. They are selected by a purely morphological criterion: among large spiral galaxies in the Local Group, those with maximum morphological elongation (minimum axis ratio b/a) are the strongest geometric pointers to the S3 torsion field.

Galaxy	Type	Inclination	Axis Ratio b/a	SEC Status
M31 (Andromeda)	SA(s)b	77° (near edge-on)	≈ 0.27	MAXIMUM — strongest pointer
M33 (Triangulum)	SA(s)cd	54° (moderate)	≈ 0.50	Second strongest pointer
Milky Way	SBbc	Internal observer	N/A	Excluded (internal)

9.3 Theta_0 Derivation

The duality sphere D is constructed from M31 ($l=121.2^\circ$, $b=-21.6^\circ$) and M33 ($l=133.6^\circ$, $b=-31.3^\circ$). The pole gives $l_D = 305^\circ$, $b_D = +25^\circ$. The great-circle distance to Sgr A*:

$$\theta_0 = \arccos(\cos(25^\circ) \cdot \cos(55^\circ)) = \arccos(0.5198) \approx 58.7^\circ \approx 59^\circ$$

NOT a free parameter — geometrically determined by M31/M33 coordinates alone.

9.4 Independent Determination via GCD_epsilon — OT-11 CONFIRMED (V18.0)

Given any set of observed SMBH angular distances $\{\theta_i\}$ from D , the fundamental quantum θ_0 can be determined by GCD_{ϵ} — the greatest common divisor with angular tolerance ϵ .

OT-11 RESULT (V18.0): GCD_{ϵ} converges to $\theta_0 = 58.963^\circ$ ($\Delta=0.313^\circ$) with all objects, and to $\theta_0 = 59.169^\circ$ ($\Delta=0.519^\circ$) with independent objects only (no DEF anchors). Combined with candidate coordinates: $\theta_0 = 58.653^\circ$ ($\Delta=0.003^\circ$). OT-11 is substantively confirmed — the GCD converges to the HTM prediction independent of Sgr A*.

9.5 Shell Spectrum

Shell n	θ_n	Known SMBH	Candidate	Observed θ	Δ	Independent?	Status
$n=1$	58.7°	Sgr A* (Milky Way)	—	58.7°	0.05°	DEF anchor	Confirmed
$n=2$	117.4°	NGC 1052 (AGN)	—	118.37°	0.97°	YES	Candidate
$n=3$	176.1°	M31 (Andromeda)	—	175.11°	0.99°	DEF object	DEF (not indep.)
$n=4$	234.8°	—	—	—	—	—	Search zone
$n=5$	293.5°	—	—	—	—	—	Search zone
$n=6$	352.2°	—	—	—	—	—	Near antipode

9.6 Katz Fractal Dimension Analysis — OT-6 CLARIFIED (V18.0)

FD = log(n) / log(d / L)

OT-6 CLARIFICATION (V18.0): The original Katz-FD formula is defined for time series and 1D curves, not for point clouds on a 2-sphere. Applied to non-collinear points on S², L_{sum} > d_{max} necessarily, producing negative FD values — which correctly indicate the points do NOT lie on a straight line (consistent with shell-ring geometry). The correct test requires Box-Counting FD on the full angular distribution with >100 catalog objects. With 9 current shell objects, the Box-Counting FD = 0.20 (R²=0.80) — insufficient sample size. OT-6 remains open, reformulated: apply Box-Counting FD to the full NED/HyperLeda SMBH catalog angular distribution. Expected range: 1.2 < FD < 1.8.

Falsification point: If the full NED/HyperLeda SMBH catalog yields FD outside the range 1.2 < FD < 1.8 at 95% confidence, the Matryoshka shell structure is falsified. This test is archivally executable with the full catalog.

9.7 M_{meta} — The Metageometra Mandelbrot Analogue

M_{meta} = { lambda ∈ Lambda : Tⁿ(x₀; lambda) → x* ∈ BH-node-set } where lambda is the torsion/coupling parameter space and T is the torsion map on S². The set M_{meta} is the space of all parameter values for which our universe — with its specific SMBH positions — is a stable fixed-point attractor.

Classical Mandelbrot	Metageometra M _{meta}
Parameter c ∈ C	Parameter lambda (torsion, coupling, angle)
Iteration z -> z^2 + c	Iteration x → T(x; lambda) on S ²
Bounded orbits	Orbits converge to BH nodes
Mandelbrot set M	M _{meta} : parameter space of stable cosmologies
Julia sets: structures	Resonance rings: fractal shell structure

Open Task OT-8: Numerical exploration of M_{meta} via parameter sampling. Expected: fractal boundary whose self-similarity dimension matches Box-Counting FD (OT-6).

10. Sigma_meta Reverberation Matter (SRM) — Dark Matter as Cavity Echo

Dark matter is reinterpreted not as an undiscovered particle species but as the persistent standing-wave reverberation structure of the rift cavity Sigma_meta. The mechanism is geometrically enforced by the same torsion dynamics that generates L, a0, and eta.

10.1 The Three Outputs of Sigma_meta

Output	Type	Physical Consequence	Observable
(A) Seepage flux L	DC component	Continuous slow leakage of meta-energy	Dark energy rho_DE
(B) Echo pulse f_echo	Impulse response	Amplified reflection of baryogenesis	Baryon asymmetry eta
(C) Standing modes rho_SRM	Eigenmode spectrum	Persistent reverberation structure	Dark matter halos

All three emerge from the same Sigma_meta cavity geometry. They are not independent phenomena — they are the DC component (L), the impulse response (f_echo), and the eigenmode spectrum (SRM) of the same physical object.

10.2 SRM Energy Density Field

$$\begin{aligned} \rho_{\text{SRM}}(r,t) &= \sum_n |A_n|^2 \cdot |\Psi_n(r,t)|^2 \\ \Psi_n(r,t) &= A_n \cdot \sin(n \cdot \pi \cdot r / \Delta \Sigma(r)) \cdot \cos(\omega_n \cdot t + \phi_n) \end{aligned}$$

10.3 Geometric Origin of Dark Matter Halos

$$\rho_{\text{SRM}}(r) = \rho_0 \cdot [1 + \gamma \cdot (1 - D_f/2) \cdot (r/r_s)]^{-2/(2-D_{f,\text{eff}})}$$

with $D_{f,\text{eff}} = 0.34$ (analytically derived, Chapter 8) and $r_s = c/(2\pi \cdot a_0) = t_0 \cdot c/(2\pi)$. The exponent $-2/(2-D_{f,\text{eff}}) = -2/(2-0.34) = -2/1.66 \approx -1.205$ is close to but distinct from the NFW inner slope of -1 . This is a falsifiable deviation testable with JWST weak lensing.

10.4 SRM Scale Radius

$$r_s = c^2 / (2\pi \cdot a_0) = (2.998 \times 10^8)^2 / (2\pi \times 1.097 \times 10^{-10}) \approx 1.306 \times 10^{23} \text{ m} \approx 42.3 \text{ kpc}$$

The natural SRM scale radius of ~42 kpc is consistent with observed dark matter halo transition scales in Milky Way-mass galaxies. This is not a fitted parameter — it follows from a0 and c alone.

10.5 Connection to the RAR

$$g_{\text{obs}} = g_{\text{bar}} / (1 - \exp(-\sqrt{g_{\text{bar}} / a_0}))$$

For $g_{\text{bar}} \gg a_0$: $g_{\text{obs}} \approx g_{\text{bar}}$ (Newtonian). For $g_{\text{bar}} \ll a_0$: $g_{\text{obs}} \approx \sqrt{g_{\text{bar}} \cdot a_0}$ (SRM-dominated). This reproduces MOND phenomenology without invoking modified gravity — the physics is standard GR plus the SRM cavity contribution.

10.6 SRM Prediction Table

Prediction	Formula/Value	Test
SRM halo inner slope	$-2/(2-D_{f,\text{eff}}) \approx -1.205$ vs NFW (-1)	JWST weak lensing
SRM scale radius	$c^2/(2\pi \cdot a_0) \approx 42.3 \text{ kpc}$	MW halo turnover
No SRM in high-g env.	$\rho_{\text{SRM}} \rightarrow 0$ when $g_{\text{bar}} \gg a_0$	High-sigma clusters
SRM halo redshift evol.	$\rho_{\text{SRM}}(z) \sim a_0(z)$ — ~4x lower at $z=2$	JWST high-z rotation curves

11. Fractal Noether Dissipation and the Gamma-Functional

11.1 Extension Beyond Classical Noether

Noether's theorem requires the symmetry group to be a Lie group — continuous and differentiable. The fractal state-time on Tier 3 is NOT a Lie group. Standard Noether conservation therefore does not apply. The Fractal Noether Equation provides the correct generalisation:

$$dE/dt = -\gamma \cdot E^{(D_f/2)} \text{ with } \gamma > 0, \ 0 < D_f \leq 2$$

This is power-law decay following from the self-similarity of the symmetry structure — not exponential decay (which follows from symmetry breaking).

$$E(t) = E_0 \cdot [1 + \gamma \cdot (1 - D_f/2) \cdot t]^{(-2/(2-D_f))} \text{ for } D_f \neq 2$$

11.2 D_f per Temporal Tier

Time-Form / Tier	D_f (effective)	Physical Consequence
Torsion-time (Tier 1)	D_f = 2	Classical Noether conservation
Anti-time ~_M (Tier 2)	D_f ≈ 1.8	Power-law dissipation → L; w(z) drift; SRM anchoring
Fractal time MM (Tier 3)	D_f,geo ≈ 0.77	Shell quantisation; spatial fractal
Cross-tier coupling (2×3)	D_f,eff ≈ 0.34	f_echo amplification → eta; SRM halo profile

11.3 The w(z) Drift

$$w(z) = -1 + (1 - D_f/2) \cdot \Delta_w \cdot (1+z)^{(3 \cdot \Delta_w)} \text{ [} D_f \approx 1.8: w(z=0) \approx -0.97 \text{]}$$

HTM predicts a power-law w(z) profile in (1+z), NOT the linear CPL parametrisation w0 + wa·z/(1+z). A confirmed CPL fit in DESI DR3 would disfavour HTM.

11.4 The Gamma-Functional — Universal Domain-Agnostic Framework

$$\Gamma(t) = E_1(t) + E_2(t) + E_{\text{watch}}(t)$$

$$d\Gamma/dt = -L_{\text{eff}} \cdot (E_1 \cdot E_2)^{(D_f/2)} / E_{\text{total}}$$

Domain	E1	E2	Predicts
Cosmology	rho_DE (vacuum)	a0 (kinematics)	rho_DE, a0, eta, SRM, H0
Neuroscience	Synaptic activity	Metabolic flux	Coherence, regime-shift
Economics	Capital flow	Information flow	Crisis probability
AI / Grid	Token energy	Meta-coherence	Overfitting boundary
Distributed Systems	Shannon entropy	Meta-coherence	Cascade integrity
Galaxy dynamics	Baryonic density	SRM density	Rotation curves + SRM halos

12. SMBHs as Torsion Crossing Points — The Stabilisation Mechanism

12.1 Core Statement

Supermassive black holes do not stabilise the rift as an additional mechanism. They ARE the rift's torsion geometry made visible at its crossing points. The SMBH is the local condensation of torsion energy where shell intersection forces are maximal. It is the crossing point that creates the black hole — not the black hole that occupies the crossing point.

12.2 Why Crossing Points Produce SMBHs

The shell crossing points $\theta_n = n \times 58.7^\circ$ are the locations where the torsion energy density of Σ_{meta} reaches local maxima. At these points, the torsion geometry of the S3 fractal structure intersects — creating nodes of maximum curvature. Matter at these nodes collapses into the only stable configuration at a geometric fixed point: a black hole. The mass of the black hole IS the materialised torsion energy of the crossing.

12.3 The Stability Chain

Crossing point geometry → SMBH forms → SMBH locks crossing point → Σ_{meta} stabilised at θ_n → L regulated → expansion controlled → universe avoids heat death. If a shell position θ_n has no SMBH, the rift is locally unanchored. L leaks more freely at that angular position.

12.4 Complete Torsion Budget of Σ_{meta}

Component	Source	Effect on Σ_{meta}	Measurable via
$\tau_{sectors}$	Counter-rotation of dual sectors	Primary compression; creates L	HTM master formula
τ_{SMBH}	SMBHs at θ_n crossing points	Stabilises rift; prevents runaway	Shell spectrum OT-5
τ_{extern}	Neighbouring rifts in Megavacuum	Secondary modulation of L	H0 irregularities OT-24

13. The Megavacuum Hierarchy — V16.0

The fractal Matryoshka structure extends above our universe. Our Big Bang was a rift in ONE vacuum shell — not the only one, not the first.

13.1 The Vacuum Shell as Active Boundary

- **COMPRESSION:** The shell exerts torsion pressure on our rift from outside — this is tau that compresses Sigma_meta and regulates L.
- **EXPANSION PERMISSION:** The shell simultaneously yields space into which our universe expands. Its yield rate L_meta permits our expansion into the Megavacuum.

$$\rho_{DE_total} = L_internal + L_meta$$

This two-component structure of dark energy may explain why the Hubble Tension persists even after the Fractal Scale Bias correction.

13.2 External Torsion Forces — tau_extrn

$$\tau_{total} = \tau_{internal} + \tau_{SMBH} + \sum_i \tau_{extrn,i}$$

OT-24 Prediction: Irregular changes in H0 over cosmic time are not measurement errors. They are physical signals of tau_extrn — torsion events in neighbouring vacuum shells affecting our rift.

13.3 Fundamental Constants as Rift-Specific

Our $L = 1.386e-44$ kg/m³/s is LOCAL — specific to our rift. Other rifts in other shells have different L values and therefore different physical constants, different a0, different eta.

14. Fractal Scale Bias and the Hubble Tension

$$a_0 = c \cdot H_0 / (2\pi) \Rightarrow \Delta a_0 / a_0 = \Delta H_0 / H_0 \text{ [exact, zero free parameters]}$$

Quantity	Value	Source	Match
$\Delta H_0 / H_0$	7.67%	SH0ES vs Planck 2018	Reference
$\Delta a_0 / a_0$	8.63%	McGaugh 2016 vs HTM	Within 1% of $\Delta H_0 / H_0$
Ratio	0.962	Computed	$\leq 1\%$ discrepancy

The Hubble Tension is the same phenomenon as the a_0 residual — two measurements of the same underlying fractal scale bias. OT-17: the apparent $w < -1$ at $z \approx 0.5$ (DESI DR2) is an echo reflection from the Tier-2 anti-time layer, not a genuine phantom crossing.

14.1 $a_0(z)$ as Cosmic Clock

Redshift z	$t(z)$ [Gyr]	$a_0(z)$ [$1e-10$ m/s ²]	vs Local	Observable via
0	13.8	1.097	reference	+100% SPARC local
0.5	8.6	1.76	+60%	SPARC extended
1.0	5.9	2.56	+133%	KMOS galaxy surveys
2.0	3.3	4.57	+316%	JWST high- z
3.0	2.2	6.86	+525%	JWST + ELT (future)

Falsification condition: if a_0 is measured constant at $z = 1-2$ within 10% scatter (JWST data), HTM is falsified.

15. Additional Structural Predictions

15.1 Precession Cycle as Tier-3 Resonance — OT-20

$T_{\text{prec}} = 25,771.57$ years is the galactic fundamental resonance tact of Tier-3. Sgr A* is the $n=1$ shell crossing point at $\theta_0 = 58.7^\circ$. The global torsion field enforces periodic return of the Earth axis — the observable expression of $T^n(x) = x$ on Tier 3.

15.2 Planet 9 / PBH as Torsion Shell Resonance — OT-21

$$T_{\text{prec}} / T_{\text{P9}} = 25,771.57 / 5,000 \approx 5.154 \approx 5:1$$

HTM parallel to Einstein-Mercury: anomalous TNO clustering explained by local torsion field enforcing shell resonance positions. Candidate HTM-OT21-A identified at RA 03h 31m 59.91s, Dec +09° 12' 35.4" — unidentified infrared source requiring follow-up.

16. Lagrangian Formulation

L_total = L_EH + L_Torsion + L_Info + L_Coupling

L_EH = (R - 2Lambda) / (16pi.G) [Einstein-Hilbert]

L_Torsion = alpha . T_munu_rho . T^munu_rho [torsion kinetic term]

L_Info = beta.(dI/dt)^2 + gamma.(dI/dlambda)^2 [information field]

L_Coupling = gamma_c . (J_A.J_D + J_B.J_C) [angular momentum cross-coupling]

16.1 Modified Friedmann Equations (FLRW Limit)

H2 = (8pi.G/3)(rho_m + rho_DE + rho_SRM) + (8pi.G/3).rho_Torsion - k/a^2

rho_Torsion = alpha . T2 = 6alpha . H2 [T = 6H2 in FLRW]

rho_SRM = Sum_n |A_n|^2 . |Psi_n|^2

16.2 Theoretical Lineage

Thinker	Contribution	HTM Connection
Boltzmann	Statistical mechanics; entropy	S3 statistical structurethernet
Gibbs	Thermodynamic ensembles	Ensemble of dual sectors
Noether (1915)	Symmetry-conservation duality	FND extends to fractal symmetries
Shannon (1948)	Information as physical quantity	L_Info; information as primordial substrate
Mandelbrot (1975)	Fractal geometry	D_f, shell structure, Tier-3, M_meta
McGaugh et al. 2016	RAR empirically	a0 prediction and FSB
DESI 2024/2025	w ≠ -1 at >3sigma	w(z) confirmed; FSB support

17. OT-5 — Statistical Evaluation of the Shell Spectrum (V18.0)

17.1 Catalog and Methodology

93 objects with secure dynamical SMBH mass measurements compiled from: Graham (2008) — 50 direct M_{BH} measurements; McConnell & Ma (2013) — 72 dynamical; van den Bosch (2016) MASSIVE Survey; Thater et al. (2019-2023) — SINFONI/MUSE; ALMA/WISDOM Kabasares (2023/2024). Local Group definers (Sgr A*, M31, M33) and scaling-relation masses (M_{sigma} , M_{L}) excluded.

$$\begin{aligned} \text{Distance metric: } \cos(c) &= \sin(b_D)\sin(b) + \cos(b_D)\cos(b)\cos(l_D - l) \mid \\ \text{D-pole} &= (l=305^\circ, b=+25^\circ) \end{aligned}$$

Verification: NGC 1052 (J2000 RA=02h41m04.8s, Dec=−08°15'21") → $l=200.77^\circ$, $b=-55.40^\circ$ → $c=118.32^\circ$. Shell $n=2 = 117.30^\circ$, $\Delta=1.02^\circ < 5^\circ$. Confirmed independent hit.

17.2 Statistical Results

Test	Result	p-value	Interpretation
Shell hit count	8/93 (expected 14.7)	—	Deficit — not excess
Binomial (one-sided)	$k=8$, $p_{\text{hit}}=15.8\%$	0.986	Not significant
Monte Carlo (100k)	mean=14.7, sd=3.4	0.985	Not significant
KS-test vs. isotropy	$D=0.191$	0.002	Survey bias — see 17.3
Pole-shift (20 rand.)	p range 0.00–0.13, mean=0.008	—	KS not D-pole specific
Spin alternation (3/3)	3 confirmed	0.125	Not yet significant — see 17.4

17.3 KS Result and Survey Bias

The KS $p=0.002$ reflects non-isotropy of the catalog itself — not the HTM signal. Maximum CDF deviation occurs at $\theta=65.5^\circ$ (Virgo cluster concentration). Only 38.7% of objects lie at $d>90^\circ$ (isotropic expectation: 50%). The pole-shift test confirms: 20 random pole locations yield similar KS significance (mean $p=0.008$). The KS signal is not specific to the D-pole.

17.4 The Gear Mechanism — Novel Independent Spin Prediction (V18.0)

Shell-SMBHs function as a cosmic gear mechanism stabilising Σ_{meta} . Each holds the torsion rift stable against collapse or runaway expansion. Adjacent shell objects must rotate in opposite directions to maintain torsion equilibrium — alternating prograde/retrograde spin orientation:

Shell n	Predicted group	Spin direction	Known object	Observed	Source	Status
n=1	A	prograde	Sgr A*	prograde	EHT 2022	Confirmed
n=2	B	retrograde	NGC 1052	retrograde	Baczko et al. 2016	Confirmed
n=3	A	prograde	NGC 0315	prograde	Daly 2023	Confirmed
n=4	B	retrograde	Unknown (K2-3?)	—	—	Predicted
n=5	A	prograde	Unknown	—	—	Predicted
n=6	B	retrograde	Unknown	—	—	Predicted

Binomial test: $P(3/3 \text{ correct at } p=0.5) = 0.125$. Not yet significant. Requires $n \geq 6$ for $p < 0.05$. Our Galaxy (Sgr A, Shell $n=1$, prograde) is Gear Group A. NGC 1052 (Shell $n=2$, retrograde) is our nearest confirmed gear partner.*

17.5 Shell Candidates — NGC 3338 and NGC 3370 (OT-29 CONFIRMED V18.0)

NGC 3338 (K2-1): Barred spiral, Leo, ~74 Mly. Diameter >110,000 ly. Delta from Shell n=2: 1.5°. No dynamical BH mass exists. Predicted spin: retrograde. Recommended: ALMA CO(2-1) molecular gas dynamics. (OT-26)

NGC 3370 (K2-2, 'Silverado Galaxy'): Sc spiral, Leo, ~82 Mly. Comparable to Milky Way. Delta from Shell n=2: 2.1°. No dynamical BH mass exists. Predicted spin: retrograde. Recommended: VLT/SINFONI stellar kinematics. (OT-27)

Retrograde candidates (Group B — Shell n=2,4,6):

Designation	l (gal)	b (gal)	RA (J2000)	Dec (J2000)	Predicted spin
K2-1 / NGC 3338	105.4°	+34.9°	10h42m	+13°44'	retrograde
K2-2 / NGC 3370	117.7°	+37.3°	10h47m	+17°16'	retrograde
K2-3	55.5°	−82.9°	18h38m	−33°39'	retrograde
K2-4	68.5°	−85.7°	18h39m	−30°33'	retrograde
K2-5	93.2°	+29.8°	10h10m	−09°01'	retrograde

Prograde candidates (Group A — Shell n=1,3,5):

Designation	l (gal)	b (gal)	RA (J2000)	Dec (J2000)	Predicted spin
K1-1	282.0°	−29.3°	22h29m	+03°04'	prograde
K1-2	327.0°	−29.3°	23h36m	−33°30'	prograde
K1-3	297.3°	−33.2°	22h42m	−10°12'	prograde
K1-4	343.0°	−20.5°	00h39m	−44°09'	prograde
K1-5	312.7°	−33.2°	23h02m	−22°03'	prograde

18. External Convergence: Mosher kappa-Framework

Jason Mosher's kappa-framework (*Emergent Space and Time: A Physically Motivated First-Principles Unification*, Zenodo v5, DOI: 10.5281/zenodo.19816994, April 2026) derives the MOND acceleration scale from a single measured atomic clock constant $\kappa = 1.242 \times 10^{-54} \text{ m}$. His derivation yields:

$$a_0 = c \cdot H_0 / (2 \cdot \pi) = 1.042 \times 10^{-10} \text{ m/s}^2 \text{ [Mosher kappa-framework]}$$

Metageometra V6.1 derives the same result from S3 resonance ring topology:

$$a_0 = c / (2 \cdot \pi \cdot t_0) = 1.097 \times 10^{-10} \text{ m/s}^2 \text{ [Metageometra HTM]}$$

Two completely independent starting points — Mosher's scalar proton sum and per-atom tick constant κ from atomic clock measurements vs. Metageometra's S3 topology — converge on the same acceleration scale $a_0 = c \cdot H_0 / (2 \cdot \pi)$. This independent convergence constitutes external evidence that the $2 \cdot \pi$ geometric factor is physically real.

Mosher's observation that the two derivations diverge by 8% (global topology vs. local enumeration) is incorporated in the V20.1 addendum. The factor $2 \cdot \pi$ appears in Mosher's framework as the geometry of a circular orbit against a linear expansion rate — structurally identical to its role in Metageometra as the S3 resonance ring circumference.

Citation note: Jason Mosher (@JasonPaulMosher) has given explicit permission for citation of his paper and his observation regarding global topology vs. local enumeration (Twitter/X exchange, 28 April 2026).

19. Statistical Tests — V20.0 Status

Test	Result	p-value	Interpretation
OT-37: IFS in R4	D_H=0.77, tetrahedron geometry confirmed	—	CONFIRMED
OT-38: Shell formula scan	delta=1.0, chi=59.1, residual=0.609	—	CONFIRMED
OT-39: V20 vs 97 SMBHs	17/97 hits (expected 12.2)	0.10	Trend, not yet significant
OT-40: Perm. test vs CF3	z = 3.62 sigma	0.0009	Significant — CF3 caveat
OT-00: Master Formula	rho_DE, a0, eta from L	—	CONFIRMED parameterfree
OT-02: f_echo Path B	N=274.9 (<1% deviation)	—	CONFIRMED
OT-02: f_echo Path E	eta=5.58e-10 (8% deviation)	—	PRIMARY RESULT V20
OT-11: GCD_eps	theta_0 = 58.963° (Delta=0.313°)	—	CONFIRMED
OT-18: Torsion shock SPARKS	RC_KS=0.95	0.95	No signal — upper bound set
OT-30: GW clustering	p=0.58 (n=15 too small)	0.58	Inconclusive

Note on OT-40: The CF3 catalog covers only $\theta_D = 0-45^\circ$, creating a survey bias that may inflate the signal. The $p=0.0009$ result must be validated with a full-sky catalog (2MRS, SDSS) before it can be treated as confirmed.

20. Falsifiability Matrix — V20.0

#	Prediction	Instrument	Falsified If
1	$w(z)$ power-law drift, $w > -1$, non-CPL	DESI DR3 2026-2027	CPL at 5 sigma
2	$a_0(z) \sim 4x$ at $z=2$ — STRONGEST	JWST+ELT 2025-2030	a_0 const within 10%
3	SMBH shell spectrum excess at $n \sim 59 \pm 5^\circ$	NED/HyperLeda	Isotropic at 95% CL
4	SRM halo inner slope -1.205 (not -1)	JWST lensing	NFW slope at >3 sigma
5	Katz-FD (Box-Counting) BH: $1.2 < FD < 1.8$	NED/HyperLeda	FD outside range
6	SEC: spiral arm persistence $\sim \theta_n$	SPARC+GAIA	No correlation
7	Cosmic voids avoid θ_n ; filaments align	SDSS/DES	Random void distribution
8	SMBH mass gradient: outer shells larger	NED catalog	No mass-shell correlation
9	SMBH clustering at $\theta \sim 90 \pm 2^\circ$	97-SMBH catalog	No excess at 90°
10	Backward shell $n=4$: NGC 2273, NGC 6251	Kinematic obs.	Retrograde not confirmed
11	Shell $n=2$ spin: NGC 3338 retrograde (V18)	ALMA/VLT 2026+	Prograde measurement

21. Open Tasks and Future Work

#	Task	Status	P
OT-1	Analytic $D_{f,diss} = 1/(n \cdot D_{f,geo})$ from tau-isotropy	CLOSED V8.1	*****
OT-2	f_{echo} from first principles: verify $f_{echo} = \exp(D_{f,eff} \cdot \ln(\dots))$	Open	****
OT-3	Numerical confirmation $G_{eff} = G/(2c^2)$ from L_{total} variation	Open	***
OT-4	α from Milky Way rotation curve $\rightarrow$ verify $k \sim O(1)$	Open	***
OT-5	KS-test $\theta_{0,0}$ shell spectrum vs full NED/HyperLeda SMBH catalog	EVALUATED V18.0 — Ch.17	****
OT-6	Katz-FD: method clarified; Box-Counting FD requires >100 objects from NED	REFORMULATED V18.0	****
OT-7	$w(z)$ power-law curve from $D_f=1.8$ vs DESI DR2/DR3 data	Open	****
OT-8	Numerical exploration of M_{meta} (Metageometra Mandelbrot set)	Open	***
OT-9	C_{Zeit} : monoidal category? Functor to Topos?	Open	**
OT-10	Delta-theta from Megavacuum initial conditions — full analytical derivation	Open	***
OT-11	GCD_eps: verify $\theta_{0,0}$ from full SMBH angle catalog independently	CONFIRMED V18.0: 58.963°	****
OT-12	Gamma-Functional regime-shift detection in power grid cascade models	Open	**
OT-13	SRM eigenmode spectrum: numerical computation of $ A_n ^2$ from cavity geometry	Open	****
OT-14	SRM halo profile: fit to observed galaxy rotation curves (SPARC sample)	Open	****
OT-15	SRM + baryonic feedback: derive $\delta_{\Sigma}(r)$ from torsion compression tau	Open	***
OT-16	$\delta_{a0}/a_0 = \delta_{H0}/H_0$ — FSB	RESOLVED V10.0	*****
OT-17	Phantom Crossing as anti-time echo — projection operator formalism	Open	*****
OT-18	KS-test: RAR residuals vs Delta-theta_shell	Open	****
OT-19	SMBH shell angle $\theta_{0,0}(z)$ at high z via JWST	Open	***
OT-20	Precession cycle T_{prec} as Tier-3 resonance — formal proof	Open	****
OT-21	Planet 9/PBH as 5th sub-harmonic; HTM-OT21-A follow-up	Open	*****
OT-22	SEC formal proof from S3 field equations	Open V14.0	*****
OT-23	Cosmic voids at non- θ_n positions; filaments at θ_n — KS-test SDSS	Open V16.0	*****
OT-24	H_0 irregularities over cosmic time as τ_{extern} signal from Megavacuum	Open V16.0	****
OT-25	L_{meta} : formal derivation of vacuum shell yield rate; relation to dark energy	Open V16.0	*****
OT-26	NGC 3338: ALMA CO(2-1) spin measurement — retrograde predicted (NEW V18)	NEW V18.0	*****
OT-27	NGC 3370: VLT/SINFONI stellar kinematics spin measurement (NEW V18)	NEW V18.0	*****
OT-28	Extend spin-alternation test to $n \geq 6$ confirmed objects for $p < 0.05$ (NEW V18)	NEW V18.0	*****
OT-29	GCD_eps on K2-1..K2-5 and K1-1..K1-5 candidates	CONFIRMED V18.0: $\Delta < 0.01^\circ$	****
OT-37	IFS in R4 $\rightarrow$ bisected tesseract geometry	CONFIRMED V20.0	*****
OT-38	Unique shell formula scan: $\delta=1.0^\circ$, $\chi=59.1^\circ$	CONFIRMED V20.0	*****
OT-39	V20 shell predictions vs 97-SMBH catalog ($\pm 2^\circ$)	Open V20.0	****
OT-40	Permutation test vs CF3 catalog — full-sky validation needed	Open V20.0 — caveat	****

22. Key Formulas Compact — V20.0

Shell angle: $\theta_n = \arccos(\cos(n * 1.0^\circ) * \cos(n * 59.1^\circ))$

Fractal dimension: $D_{f,geo} = \ln(2) / \ln(1/0.406) = 0.769$

Echo amplitude: $f_{echo} = \exp(0.3388 * 274) = 2.07 \times 10^{40}$

MOND scale: $a_0 = c / (2\pi t_0) = 1.096 \times 10^{-10} \text{ m/s}^2$

Seepage rate: $L = \rho_{DE} / t_0 = 1.386 \times 10^{-44} \text{ kg/m}^3/\text{s}$

Baryon asymmetry Path E (PRIMARY – V20.0):

$\eta = (1 - \cos(\chi)) / (2 - \cos(\chi)) / (2\pi^2)^{(6 + \ln 2 / \ln(1/\rho))}$

$= 5.58 \times 10^{-10}$ (8% deviation from $\eta_{obs} = 6.104 \times 10^{-10}$ – zero free parameters)

Torsion-Isotropy Theorem: $D_{f,diss} = 1/(3D_{f,geo})$

SRM halo slope: $-2/(2 - D_{f,eff}) = -1.205$

SRM scale radius: $r_s = c^2/(2\pi a_0) \approx 42.3 \text{ kpc}$

Appendix C — Task Evolution, Failures, and Emergent Corrections

This appendix documents open tasks that failed, were methodologically corrected, or changed fundamentally as a result of emergent framework dynamics. It constitutes evidence that Metageometra is self-correcting rather than ad-hoc fitted. Every failure listed here was discovered internally — not by external criticism.

C.1 Geometric Foundation: Cube $\rightarrow S^3 \rightarrow$ Tesseract

Original state (v0.9–V19): The framework used a cube as an informal visualisation aid for the diagonal cross-offset (Kreuzversatz) of torsion forces, then formalised the geometry as an abstract closed 3-sphere S^3 . The shell formula was purely linear: $\theta_n = n \times 58.7^\circ$.

Problem that emerged: The linear formula broke down above $n=3$. Shells $n=4,5,6$ exceeded 180° and had no physical interpretation — V19 simply had no shells above the Duality Sphere. Predicted search zones remained observationally empty. Cen A at 8.1° was unexplained substructure that fit nowhere.

Resolution (V20.0, OT-37/38): The IFS attractor constraint in R^4 mathematically forced the geometry to a bisected tesseract — not a choice, but a consequence of $N_{\text{sub}}=2$ and $\rho=0.406$. The new formula $\theta_n = \arccos(\cos(n \cdot \delta) \cdot \cos(n \cdot \chi))$ folds back automatically via $\arccos$, producing backward shells $n=4,5,6$. Cen A at $n=6$ (8.1°) is now geometrically necessary. The cube of v0.9 was the first intuitive shadow of what the tesseract would become.

C.2 OT-38b: Geometric f_{echo} Formula — FAILED

Hypothesis tested: Derive f_{echo} directly from tesseract geometry:

$$f_{\text{echo_geo}} = ((1 - \cos(\delta)) / (\rho^2 \cdot \sin(\chi)))^{D_{\text{eff}}}$$

Numerical result (computed): Argument = $0.00108 \ll 1$. Result: $f_{\text{echo_geo}} = 0.099$ ($\log_{10} = -1.01$). Required: $f_{\text{echo}} = 2.65 \times 10^{40}$ ($\log_{10} = +40.42$). Error: 41 orders of magnitude.

Diagnosis: The geometry ratio $(1 - \cos(\delta)) / (\rho^2 \cdot \sin(\chi))$ describes spatial compression — inherently $\ll 1$. The thermodynamic depth $N = \ln(\rho_{\text{Planck}} / \rho_{\text{DE}}) \approx 274$ is orthogonal — inherently $\gg 1$. These two quantities cannot be combined in a simple power law. The formula conflated two physically orthogonal regimes.

Consequence: OT-38b is closed as FAILED. The FND thermodynamic path $f_{\text{echo}} = \exp(D_{f,\text{eff}} \times N)$ remains the sole canonical derivation of f_{echo} . Tesseract geometry determines the shell angles and baryon asymmetry (Path E), but not f_{echo} .

C.3 OT-6: Katz Fractal Dimension — Method FAILED, Reformulated

Original task (v5.1): Apply Katz FD = $\log(n) / \log(d/L)$ to SMBH angular positions on S^2 to test shell structure.

Problem discovered (V18.0): Katz-FD is defined for 1D time series and curves. On a 2-sphere, non-collinear points necessarily have $L_{\text{sum}} > d_{\text{max}}$, producing negative FD — mathematically meaningless as a fractal dimension.

Resolution: Reformulated to Box-Counting FD on the full NED/HyperLeda catalog (>100 objects). Expected range: $1.2 < \text{FD} < 1.8$. Task open but correctly specified.

C.4 OT-18: RAR Shell Proximity Test — Retroactively a Predicted Null

Original framing: Test whether SPARC galaxies near HTM shells show different RAR scatter than galaxies far from shells.

Problem discovered via pre-registration check: $L = \rho_{DE}/t_0$ is defined as a strictly global quantity — one number per universe/rift, with no shell-local variation anywhere in the framework. The test was therefore a predicted null from the outset.

Status: OT-18 null result ($p_{KS} = 0.95$) is fully consistent with the framework. It sets an upper bound on hypothetical torsion-shock amplitude $|\delta L/L|$. The genuinely falsifying test is $a_0(z)$ via JWST — not shell-proximity RAR scatter.

C.5 OT-39/40: V20 Shell Formula vs SMBH Catalog — Current Status

Shell	theta_pred	Hits (±2°)	Key objects (OT-39, N=97)
n=1	59.11°	9	NGC 3489, NGC 3384, NGC 3379, Sgr A*, NGC 3377, NGC 3842, NGC 3608, M85, NGC 4350
n=2	118.18°	5	UGC 3789, NGC 2748, NGC 1600, NGC 1300, NGC 4291
n=3	175.96°	4	M32, M31, NGC 404, NGC 315
n=4	123.51°	4	NGC 2273, NGC 6251, NGC 1550, NGC 1052
n=5	64.60°	4	NGC 4889, NGC 2960, NGC 4278, NGC 3227
n=6	8.07°	2	NGC 5128 (Cen A), NGC 4751

Total: 28/97 hits. OT-40 permutation test vs CF3 background (100k samples): $T_{obs}=17.47$, T_{rand} mean=8.90, $z=3.62$ sigma, $p=0.0009$. Shell n=6 shows no CF3 signal ($z=-3.17$, $p=1.0$) — expected, as CF3 covers only $\theta_D=0.5^\circ-45^\circ$. Full-sky validation (2MRS/SDSS) required before treating OT-40 as confirmed.

C.6 D_f Discrepancy — Seven Versions of Silent Contradiction

Silent inconsistency (v0.2–V7.0): $D_f = 0.77$ was used everywhere. f_{echo} requires $D_f \approx 0.34$. A factor-of-two contradiction existed for seven major versions without detection — because f_{echo} and $D_{f,geo}$ were never compared explicitly in the same derivation chain.

Resolution (V8.0/8.1): Torsion-Isotropy Theorem proved $D_{f,diss} = 1/(3 \cdot D_{f,geo})$ analytically from isotropic load distribution. $D_{f,eff} = 0.34$ is a geometric consequence, not a patch. The 0.77 was always correct for the geometric sector — just incomplete as the only fractal dimension.

Appendix A — Numerical Reference Table V20.0

Symbol	Value	Source / Derivation
H0 (CMB)	67.4 km/s/Mpc	Planck 2018
H0 (local)	73.0 km/s/Mpc	SH0ES / H0DN 2026
t0	4.352×1.e+17 s	1/H0
c	2.998×1.e+8 m/s	Physical constant
rho_DE	6.034e-27 kg/m3	Planck 2018
L	1.386e-44 kg/m3/s	rho_DE/t0 (def.)
a0 (HTM)	1.097×1e-10 m/s2	c/(2pi·t0)
a0 (obs.)	1.20×1e-10 m/s2	McGaugh et al. 2016
a0 (Mosher)	1.042×1e-10 m/s2	c·H0/(2pi), kappa-framework
rho (IFS)	0.406	Tesseract cell projection (OT-37)
D_f,geo	0.769	ln(2)/ln(1/rho)
D_f,diss	0.433	1/(3·D_f,geo) — Torsion-Isotropy Theorem
D_f,eff	0.3388	D_f,geo × D_f,diss
chi	59.1°	OT-38 scan — shell formula parameter
delta	1.0°	OT-38 scan — shell formula parameter
Va	0.4865	1 – cos(chi) — asymmetric tesseract fraction
eta (Path E)	5.58e-10	Va/(1+Va)/(2*pi2)^(6+D_f,geo) — V20 primary
eta (obs.)	6.104×1.e-10	Planck 2018 / BBN
f_echo	2.07×1.e+40	exp(D_f,eff × 274)
theta_0	58.65°	arccos(cos25°·cos55°) — M31/M33 geometry
r_s (SRM)	42.3 kpc	c2/(2pi*a0)
SRM slope	–1.205	–2/(2–D_f,eff)
kappa (Mosher)	1.242×1.e-54 m	Per-atom tick constant, atomic clocks
l_D, b_D	305°, +25°	M31/M33 great-circle pole — Duality sphere D
G_eff	G/(2c2)	Weak-field limit from L_total variation
alpha (FSB)	0.7433	(3–D_f,geo)/3 — Fractal Scale Bias exponent
T_prec	25,771.57 years	Bretagnon 2003; NASA — OT-20
NGC 3338	RA 10h42m, Dec +13°44'	NED; V18.0 OT-26 — K2-1 / Shell n=2 candidate
NGC 3370	RA 10h47m, Dec +17°16'	NED; V18.0 OT-27 — K2-2 / Shell n=2 candidate
GCD_eps result	58.963° (all), 59.169° (indep.)	V18.0 OT-11 — Independent theta_0 confirmation

Appendix B — Complete Derivation Chain (Anti-Post-Hoc Evidence)

This appendix documents the precise sequence of derivations across all versions, constituting the primary evidence that no element was introduced after observational comparison. Each step was formulated before the subsequent result was known.

Version	What was derived	What was NOT yet known at that point
v0.2	Dark matter as information-cavity echo; dual sectors; pi as turning constant	No L-coms, no eta, no theta_0
v0.8	Centripetal stabilisation geometry; BH as pi-boundary interface	No torsion formula, no S³ model
v0.9	S3 diagonal torsion; time as torsion rotation; Noether argument for M31/M33	No M31/M33, no theta_0, no Df
v4.0	S3 tau formula; Lagrangian; M31/M33 → theta	No Df closure, no 3-6-9 proof, no eta
v5.1	Node conditions; Katz-FD; M_meta; theta_0 shell spectrum; GCD_eps method	No Df(S3) proof, Df still inconsistent
v6.1	pi3(S3)=Z → n=3; Df formula; f_echo, eta, a0 parameter-free	No Df discrepancy resolution, no SRM
v7.0	C_Zeit category with morphisms T, Phi, Psi; L as single free phase	No Df, geo = Df,geo × Df,diss decomposition
V8.0/8.1	Df decomposition; Torsion-Isotropy Theorem; Df,diss = 1/(3*DfgeoSRM)	No SRM, no Megavacuum
V9.0	SRM cavity echo; hollow-chamber model; three Sigma_meta	No FSB, no Bubble Tension resolution
V10.0	FSB: delta_a0/a0 = delta_H0/H0 exactly	No Megavacuum, no tau_extern
V14.0	SEC: M31/M33 selection formally justified on morphological grounds	No retrospective justification of v4.0 choice
V16.0	SMBHs as crossing points; Megavacuum; tau_extern	L_meta formal derivation still open
V17.0	All derivation chains unified; Katz-FD, M_meta, GCD_eps, C_Zeit morphisms	One chain open
V18.0	OT-5 evaluated; OT-6 clarified; OT-11 confirmed; OT-29 confirmed	OT-26 Beauty Chain open; C_Sp338a, 8370 identified
V20.0	Tesseract as geometric foundation (OT-37); unique shell form	OT-37/438 Path Elliptical, fully symmetrical, wanted back/26d wider structure pending shell

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Document Status

This document is the complete self-contained master synthesis of the Metageometra framework as of V20.0 (28 April 2026). All content from V18.0 and V19.0 is preserved in full. Nothing has been omitted. V20.0 adds:

- Chapter 1: Tesseract as geometric foundation (OT-37/38 confirmed)
- Chapter 2: Forward/Backward wave structure — Cen A as geometrically necessary
- Chapter 3: Path E — baryon asymmetry from tesseract geometry (primary result)
- Chapter 18: Mosher external convergence — fully documented with Zenodo DOI
- Chapter 19: Updated statistical test table including OT-39/40
- Chapter 20: Updated falsifiability matrix including shell $n=4$ backward wave prediction
- Acknowledgments: Mosher critical questions acknowledged as intellectual contribution

The framework has not been peer-reviewed. It is shared for early scientific feedback and falsification testing. The author welcomes correspondence, critical analysis, and collaboration proposals.

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